Assignment 1 & 2: PC Parts Shop Website Design & Development

Student Name: [Chaudhari Priyanshu]

Enrollment Number: [2502031030036]

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Course Name: Web Technology with UI/UX

Course Code: 2010043228

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1. Introduction

This project involved the end-to-end design and development of a modern e-commerce platform specializing in high-performance computer components. The objective was to create a seamless user experience for building custom PCs, from browsing hardware to secure checkout.

The project was divided into two core phases:

1. **UI/UX Design:** Crafting a professional, high-tech interface using Figma.
2. **Web Development:** Implementing the design into a functional multi-page website using HTML5, CSS3, and JavaScript.

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2. Assignment 1: UI/UX Design & Repository

Task 1: GitHub Repository Management

A structured repository was established to manage version control and project organization.

- **Repository Name:** `[Your Enrollment Number]`
- **Structure:**
  - `/Assignment_1`: Contains Figma design screenshots and documentation.
  - `/Assignment_2`: Contains the live source code (HTML, CSS, JS).

Task 2: UI/UX Prototype Design

The "PC Parts Shop" UI was designed with a focus on "Tech-Modernism." Key design principles included:

- **Clean Grid Layout:** To display complex technical specifications clearly.
- **Responsive Design:** Optimized for desktop builders and mobile shoppers.

- **Visual Hierarchy:** Highlighting "In Stock" status and "Featured Products" to drive conversions.

Screens Designed:

1. **Home Page:** Hero section with "Build Your Dream PC" CTA.
2. **Products Listing:** Category-based filtering (GPU, CPU, RAM).
3. **Product Detail:** Deep-dive into technical specs and ratings.
4. **Shopping Cart:** Real-time quantity updates and price calculations.
5. **Checkout Flow:** Multi-step shipping and payment validation.
6. **Login/Register:** User authentication interfaces.

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3. Assignment 2: Website Development

Task 1: Technical Implementation

The Figma designs were converted into a fully functional website. The following features were implemented:

- **Dynamic Product Rendering:** Products are loaded via JavaScript from a centralized data file (`products-data.js`).
- **Stateful Shopping Cart:** Utilizes `localStorage` to persist cart items across browser sessions.
- **Authentication System:** A custom `auth.js` script manages user sessions, toggling between "Login/Register" and "User Profile" menus.
- **Responsive Header:** A template-based header (`HEADER_TEMPLATE.html`) ensures consistent navigation across all pages.
- **Form Validation:** Built-in validation for shipping and payment forms in the checkout process.

Technologies Used

- **HTML5:** Semantic structure for SEO and accessibility.
- **CSS3:** Custom styling featuring Flexbox/Grid, linear gradients, and responsive breakpoints.
- **JavaScript:** Logic for cart management, quantity toggles, and UI state changes.
- **Figma:** Initial UI/UX prototyping.
- **GitHub:** Version control and deployment.

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4. Key Functionalities

Feature	Description
Cart Logic	Calculates subtotal, tax (8%), and dynamic shipping (Free over $50).
Search/Filter	Allows users to find parts by name or category.
User State	Persistent login that changes the navigation bar layout automatically.
Checkout	A 3-step process: Shipping Info → Payment → Order Confirmation.

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5. Conclusion

The PC Parts Shop project successfully demonstrates the integration of professional UI/UX design with robust front-end development. The resulting website is a responsive, user-friendly platform capable of handling the complex data requirements of an e-commerce hardware store.